

WASP-12b

R-0231

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241212 · created 2026-07-17T03:35:58+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460656.9424 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.046
Transit depth (Rp/Rs)^2	2.35 +/- 1.4 [%]
Orbital Inclination (inc)	79.53 +/- 2.88
Airmass coefficient 1 (a1)	19.62 +/- 11.96
Airmass coefficient 2 (a2)	-3.0 +/- 0.63
Scatter in the residuals of the lightcurve fit is	59.9 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	7.5
Transit Duration (day)	0.1156 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.046 vs 0.1178 (deviation 0.77σ)
Duration fit vs published	0.1156 d vs 0.125 d (deviation 1.01σ)
Mid-time O–C	0.1 min
Depth SNR	3.3
Residual scatter	59.9 %

Light curve

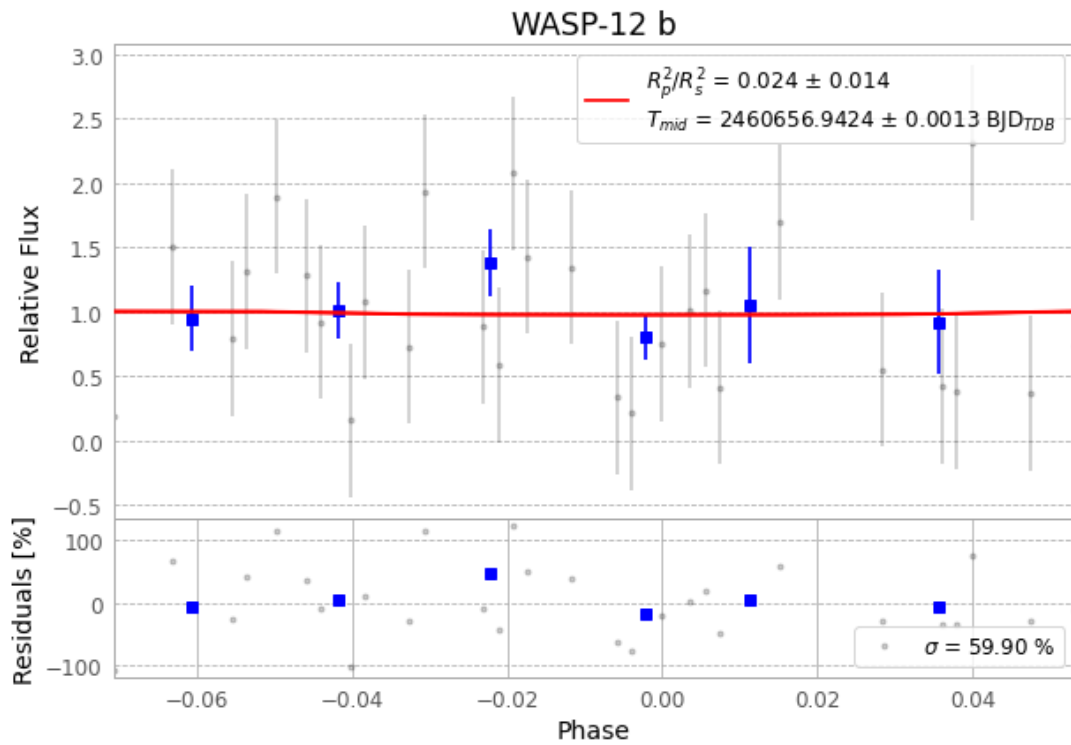


Figure 1 — FinalLightCurve_WASP-12 b_2024-12-12.png (SHA-256 67fd21e7306f438b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [540, 212], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1663fc176b8b5f47...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

72 FITS frames (47 MB), WASP-12241212082116.FITS → WASP-12241212115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241212.log (be507ea73861...), FinalLightCurve_WASP-12 b_2024-12-12.png (67fd21e7306f...), FinalLightCurve_WASP-12 b_2024-12-12.csv (f524666cb004...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 03:35Z.